

418hw2

Alex Caramanico

2023-03-06

2a

```
ymu = matrix(c(2,-3,1,-5),4,1)
sig = matrix(c(11,8,-3,-9,8,9,3,6,-3,3,2,3,-9,6,3,9),4,4,byrow=TRUE)
a = matrix(c(4,-2,1,-3),4,1)
at = t(a)
zmu = at%*ymu
zvar = at%*sig%*a
cat('z follows a normal distribution with mean',zmu,'and variance',zvar)
```

```
## z follows a normal distribution with mean 30 and variance 401
```

```
# z ~ N(30,401)
```

2b and 2c

```
# Z joint distribution: (z1 z2 z3 z4)' ---> multivariate normal, all zi normal
ymu = matrix(c(2,-3,1,-5),4,1)
sig = matrix(c(11,8,-3,-9,8,9,3,6,-3,3,2,3,-9,6,3,9),4,4,byrow=TRUE)

a1 = matrix(c(1,3,-1,-4),4,1) # for z1 linear combination
a1t = t(a1)
z1mu = a1t%*ymu # mean
z1var = a1t%*sig%*a1 # variance

a2 = matrix(c(-3,-1,1,-2),4,1) # for z2 linear combination
a2t = t(a2)
z2mu = a2t%*ymu # mean
z2var = a2t%*sig%*a2 # variance

a3 = matrix(c(1,0,1,0),4,1) # for z3 linear combination
a3t = t(a3)
z3mu = a3t%*ymu # mean
z3var = a3t%*sig%*a3 # variance

a4 = matrix(c(8,0,0,-5),4,1) # for z4 linear combination
```

```

a4t = t(a4)
z4mu = a4t*%ymu # mean
z4var = a4t*%sig*%a4 # variance

zmeanvector = rbind(z1mu,z2mu,z3mu,z4mu) # mean vector of z joint distribution
# for dispersion matrix of z joint distribution, zivar are diagonal entries
# need to find covariance between each zi, with a_i'*sig*a_j, for i,j = 1,2,3,4
covz1z2 = a1t*%sig*%a2 # Cov(z1,z2)
covz1z3 = a1t*%sig*%a3 # Cov(z1,z3)
covz1z4 = a1t*%sig*%a4 # Cov(z1,z4)
covz2z3 = a2t*%sig*%a3 # Cov(z2,z3)
covz2z4 = a2t*%sig*%a4 # Cov(z2,z4)
covz3z4 = a3t*%sig*%a4 # Cov(z3,z4)
# assemble dispersion matrix, symmetrical
zdisp = matrix(c(z1var,covz1z2,covz1z3,covz1z4,covz1z2,z2var,covz2z3,covz2z4,
                covz1z3,covz2z3,z3var,covz3z4,covz1z4,covz2z4,covz3z4,z4var),
                4,4, byrow=TRUE)
cat('2b - The joint distribution is multivariate normal with mean vector:')

```

2b - The joint distribution is multivariate normal with mean vector:

```
zmeanvector
```

```

##      [,1]
## [1,]   12
## [2,]    8
## [3,]    3
## [4,]   41

```

```
cat('and dispersion matrix:')
```

and dispersion matrix:

```
zdisp
```

```

##      [,1] [,2] [,3] [,4]
## [1,]  226 -178   66  742
## [2,] -178  110  -24 -238
## [3,]   66  -24    7   94
## [4,]  742 -238   94 1649

```

```

# 2c
# Using the information from part b, the joint distribution of z2 and z4 can
# be acquired by partitioning the original/overall joint distribution
joint2mu = matrix(c(z2mu,z4mu),2,1)
joint2disp = matrix(c(z2var,covz2z4,covz2z4,z4var),2,2,byrow=TRUE)
cat('2c - The joint distribution is multivariate normal with mean vector:')

```

2c - The joint distribution is multivariate normal with mean vector:

```
joint2mu
```

```
##      [,1]  
## [1,]    8  
## [2,]   41
```

```
cat('and dispersion matrix:')
```

```
## and dispersion matrix:
```

```
joint2disp
```

```
##      [,1] [,2]  
## [1,]  110 -238  
## [2,] -238 1649
```

3

```
mat = matrix(c(1,0,0,2),2,2,byrow=TRUE)  
solve(mat)
```

```
##      [,1] [,2]  
## [1,]    1  0.0  
## [2,]    0  0.5
```